MAT092: Remedial Course in Mathematics

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LECTURE-01

What is a Set?

Definition: A set is a collection of distinct and well-defined objects (called elements) considered as a single entity.

Intuition: We often think of sets as "bags of objects" — but mathematics demands precision. Without well-definedness, we could not even decide whether something belongs to a set.

Why well-definedness?

Consider $A = \{\text{beautiful numbers}\}$ — unclear which numbers qualify!

But $B = \{x \in \mathbb{N} : x \text{ is even}\}$ is precise.

Examples:

$S = \{1, 2, 3, 4, 5\}$

$T = \{x \in \mathbb{R} : x^2 < 2\}$

$\emptyset = \{\}$ (empty set)

Set and Membership

Notation:

$$x \in A \quad \text{means } \text{“}x \text{ is an element of } A\text{”}.$$

Examples:

$2 \in \{1,2,3\}$ but $4 \notin \{1,2,3\}$.

$\pi \in \mathbb{R}$ but $\pi \notin \mathbb{Q}$.

Remark: Sets do not care about order or repetition:

$$\{1,2,3\} = \{3,2,1\} = \{1,1,2,3,3\}.$$

Idea: Set theory is the language of mathematics — almost every structure we define later (functions, relations, numbers) is built using sets.

Relations: The Hidden Structure Between Sets

Definition: Given sets A and B, a relation R from A to B is any subset of the Cartesian product $A \times B$.

$$R \subseteq A \times B.$$

If $(a, b) \in R$, we write aRb.

Example:

$$A = B = \mathbb{N}, \quad R = \{(a, b) : a < b\}.$$

Here, $1R2$ is true but $2R1$ is false.

Motivation: Relations tell us how elements "interact" across or within sets — they capture structure, order, similarity, and more.

Types of Relations on a Set

Let A be a set and $R \subseteq A \times A$.

Reflexive: $(a, a) \in R$ for all $a \in A$. Example: $\leq$ on $\mathbb{R}$.

Symmetric: $aRb \Rightarrow bRa$. Example: "is sibling of".

Antisymmetric: aRb and $bRa \Rightarrow a = b$. Example: $\leq$ on $\mathbb{R}$.

Transitive: aRb and $bRc \Rightarrow aRc$. Example: $<$ on $\mathbb{R}$.

Special Types:

Equivalence relation: Reflexive, Symmetric, Transitive. $\rightarrow$ Partitions a set into "equivalence classes".

Partial order: Reflexive, Antisymmetric, Transitive. $\rightarrow$ Describes hierarchy or order.

Is the relation $R = \{(1,1),(2,2),(3,3),(1,2),(2,1),(2,3),(3,2)\}$ on set $A = \{1,2,3\}$ an equivalence relation?

Relations as the Foundation for Functions

Observation: Every function is a special kind of relation!

$$f \subseteq A \times B \text{ such that } \forall a \in A,\ \exists! b \in B \text{ with } (a, b) \in f.$$

Meaning: A function is a relation where each input has a unique output.

Examples:

$f(x) = x^2$ is a function from $\mathbb{R}$ to $\mathbb{R}_{\geq 0}$.

The relation $R = \{(x, y) : x^2 = y^2\}$ is not a function since $x = 2$ gives $y = \pm 2$.

Intuition: Relations give the raw "connection"; functions impose a rule of determinism.

Hidden Relations in Mathematics

Idea: Many mathematical objects secretly rely on relations.

Equality ($=$) is an equivalence relation.

Order ($\leq$) is a partial order relation.

Congruence ($\equiv \bmod n$) partitions $\mathbb{Z}$ into classes — a basis for modular arithmetic.

Parallelism in geometry, "is parallel to", is symmetric and transitive.

Moral: Relations form the invisible scaffolding of mathematics — from "equal" to "connected", "similar", or "less than".

Discord

